Technical Workshop on Curation of Climate-Smart Agricultural Advisory Content and Design of the Kenya AI Advisory Knowledge Hub

BACK-TO-OFFICE REPORT

Kenya Agricultural and Livestock Research Organization (KALRO)
KAPTAGAT ROAD, LORESHO, NAIROBI, KENYA

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Technical Workshop on Curation of Climate-Smart Agricultural Advisory Content and Design of the Kenya AI Advisory Knowledge Hub

Workstream: 1 (Sub-components 1.2 & 1.5)

Dates: 17 - 21 August 2026

Venue: KALRO Agricultural Mechanization Research Institute (AMRI), Katumani, Machakos County, Kenya

Prepared by: Kevin Kiambe, Technical Lead, ICT & Digital Agriculture, KALRO

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Approved by: Simon Mulwa, Director ICT & Digital Agriculture, KALRO

Signature _____

1. Executive Summary

KALRO, in partnership with Strathmore Agri-Food and Innovation Center (SAFIC) and supported under Workstream 1 of the Kenya AI-Enabled Agricultural Advisory Platform, convened a five-day Technical Workshop at KALRO Agricultural Mechanization Research Institute (AMRI), Katumani from 17-21 August 2026. The workshop brought together KALRO leadership, ICT and Knowledge Management teams, and subject-matter experts to curate a harmonized national advisory content inventory, agree AI-ready content and metadata standards, design the Kenya AI Advisory Knowledge Hub, and produce a stakeholder-validated roadmap for digitizing and integrating that content into KilimoSTACK and the national AI Advisory Platform.

All four expected outputs of the workshop were substantively progressed: a 28-item value-chain content inventory was harmonized and delivered; AI-ready metadata, AEZ-localization was drafted and tested; a draft Knowledge Hub design framework, including completed content for validation was produced.

This report summarizes the proceedings against the workshop programme, records key outputs, issues and lessons, and sets out recommendations to carry the work forward principally.

2. Purpose of the Workshop

The Technical Workshop was convened to initiate the curation of climate-smart agricultural advisory content and design the Kenya AI Advisory Knowledge Hub, in direct support of KilimoSTACK and the national AI Advisory Platform.

2.1 Objectives

- Develop a harmonized inventory of validated agricultural advisory content across priority crop, soil, livestock, pest/disease and climate value chains.
- Agree on standards for AI-ready agricultural knowledge representation, metadata tagging, AEZ localization and content certification.
- Design the Kenya AI Advisory Knowledge Hub, including its federation model, governance and One Health content validation.
- Produce a validated roadmap for content digitization and integration into KilimoSTACK and the national AI Advisory Platform.

2.2 Expected Outputs

- Harmonized national advisory content inventory (priority value chains).
- Agreed AI-ready metadata, AEZ and content-classification standards.
- Kenya AI Advisory Knowledge Hub design framework, including One Health validation.
- Stakeholder-validated roadmap for content digitization and integration.

3. Summary of Proceedings

3.1 Day 1 (17 August 2026): Opening, Context Setting & Launch of the Content Inventory

The workshop opened with welcome remarks from Mr. Simon Mulwa, the Director ICT and Digital Agriculture, who framed KALRO's role as the national data custodian within the SAFIC/KilimoSTACK partnership. The Mr. Victor Lang'at, Assistant Director, ICT & DA then presented the workshop objectives, expected outputs and five-day roadmap, establishing a shared understanding of the process ahead.

Mr. Kevin Kiambe followed with a session on KilimoSTACK and the National AI Advisory Platform, building a common understanding of how the week's outputs would feed the Digital Public Infrastructure (DPI). The Knowledge Management team then led participants through the Content Inventory Methodology, Templates and Value-Chain Grouping session, resulting in an agreed inventory template and the formation of value-chain working groups. The day closed with parallel group work drafting inventory entries for priority crop, soil, and pest/disease value chains.

> ***Day Output:*** *Draft inventory entries for crop, soil, and pest/disease advisory content; value-chain working groups established.*

3.2 Day 2 (18 August 2026): Completing & Harmonizing the National Advisory Content Inventory

Day 2 opened with a recap of Day 1 progress before working groups resumed curation, extending coverage to livestock, fisheries, Climate-Smart Agriculture (CSA) advisory content. A total of 28 value chains and cross-cutting topics were divided among five working groups, led by Mr. Kevin Berry Okombo, Henry Wanyama, Joan Lang'at, Mr. Richard Kedemi and Ms. Primrose Wanyama, as summarized in Table 1. System development was carried out by Mr. Kevin Kiambe and Mr. Nelson Kamau.

Table 1: Value-chain working-group assignments and content status

No.	Value Chain / Topic	Category	Working Group Lead
1	Banana	Crop	Berry
2	Avocado	Crop	Berry
3	Mango	Crop	Berry
4	Watermelon	Crop	Berry
5	Plant Pest	Cross-Cutting	Berry
6	Plant Diseases	Cross-Cutting	Berry
7	Animal Health	Livestock	Berry
8	Macadamia	Nuts	Berry
9	Dairy Goats	Livestock	Berry
10	Meat Goat	Livestock	Berry
11	Indigenous Chicken	Livestock	Henry
12	Beans	Crop	Henry
13	African Indigenous Vegetables (Pumpkin, Spider Plant, Cowpea, Amaranth)	Crop	Henry
14	Green Grams	Crop	Henry
15	Natural Resource Management (NRM)	Cross-Cutting	Henry
16	Sorghum	Crop	Joan
17	Tomato	Crop	Joan
18	Sheep	Livestock	Joan

19	Market	Cross-Cutting	Joan
20	Fisheries	Livestock	Kedemi
21	Maize	Crop	Kedemi
22	Coffee	Crop	Kedemi
23	Cassava	Crop	Kedemi
24	Potato	Crop	Primrose
25	Dairy	Livestock	Primrose
26	Millet	Crop	Primrose
27	Weather	Cross-Cutting	Primrose
28	Hides and Skins	Livestock	Primrose
	Developers		
1	System Development	Backend Development	Kiambe
2	System Development	Frontend Development	Kamau

Following curation, groups cross-checked their entries against existing KALRO and partner repositories to identify gaps, then applied agreed quality-screening criteria for scientific accuracy and currency. The afternoon plenary consolidated all group submissions into a single, de-duplicated national advisory content inventory, which was prioritized by value chain and urgency for digitization.

> ***Day Output:*** *Harmonized, validated and prioritized national advisory content inventory (Expected Output 1)*

3.3 Day 3 (19 August 2026): AI-Ready Content Standards & Metadata Framework

Day 3 focused on defining what "AI-ready" content requires. The ICT Team led participants through metadata tagging and classification schema design, followed by sessions establishing an Agro-Ecological Zone (AEZ) localization framework and seasonal/climate-context tagging conventions for advisory content. A draft protocol for version control and content update-management was also developed, alongside content certification criteria and quality standards led jointly by Knowledge Management and Subject Area Experts.

In the afternoon, participants piloted the draft metadata schema against sample entries from the harmonized inventory, testing its practical applicability before the day closed with a plenary review. The Director, ICT & DA and Assistant Director, ICT & DA presented the agreed, tested standards for AI-ready knowledge representation as Expected Output 2.

> ***Day Output:*** *Agreed and pilot-tested standards for AI-ready content, metadata, AEZ tagging and certification (Expected Output 2).*

3.4 Day 4 (20 August 2026): Design of the Kenya AI Advisory Knowledge Hub

The fourth day addressed the design of the Kenya AI Advisory Knowledge Hub itself. The Director, ICT & DA opened with the Hub's design principles and success criteria, before the ICT Team led development of a draft information architecture and content taxonomy. A session on the federation model examined how universities, research institutions and other agencies could contribute externally validated content to the Hub. Minimum coverage on the One-Health content was observed.

Day Output: *Draft Kenya AI Advisory Knowledge Hub design framework, with One Health content (Expected Output 3).*

3.5 Day 5 (21 August 2026): Digitization & Integration Roadmap, Stakeholder Validation & Closing

The final day opened with a recap of the week's outputs across the inventory, standards, and Hub design. The ICT & DA and Knowledge Management teams then led development of a content digitization roadmap, sequencing and prioritizing value chains for digitization, followed by an integration roadmap linking content to KilimoSTACK and the AI Advisory Platform and identifying technical dependencies.

A session on resourcing, timelines and institutional responsibilities produced a RACI matrix and indicative implementation timeline. All four workshop outputs, the inventory, standards, Hub design framework and roadmap, were then presented in a stakeholder validation session for wider feedback, chaired by the Assistant Director ICT & DA. Feedback was incorporated into final versions of the outputs, and a consolidated action plan with owners and follow-up dates was agreed before the workshop closed formally.

Day Output: *Stakeholder-validated roadmap for content digitization and integration (Expected Output 4), plus a signed action plan closing the workshop.*

4. Key Outputs Achieved

Table 2 summarizes progress against each of the workshop's four expected outputs.

Table 2: Status of expected outputs

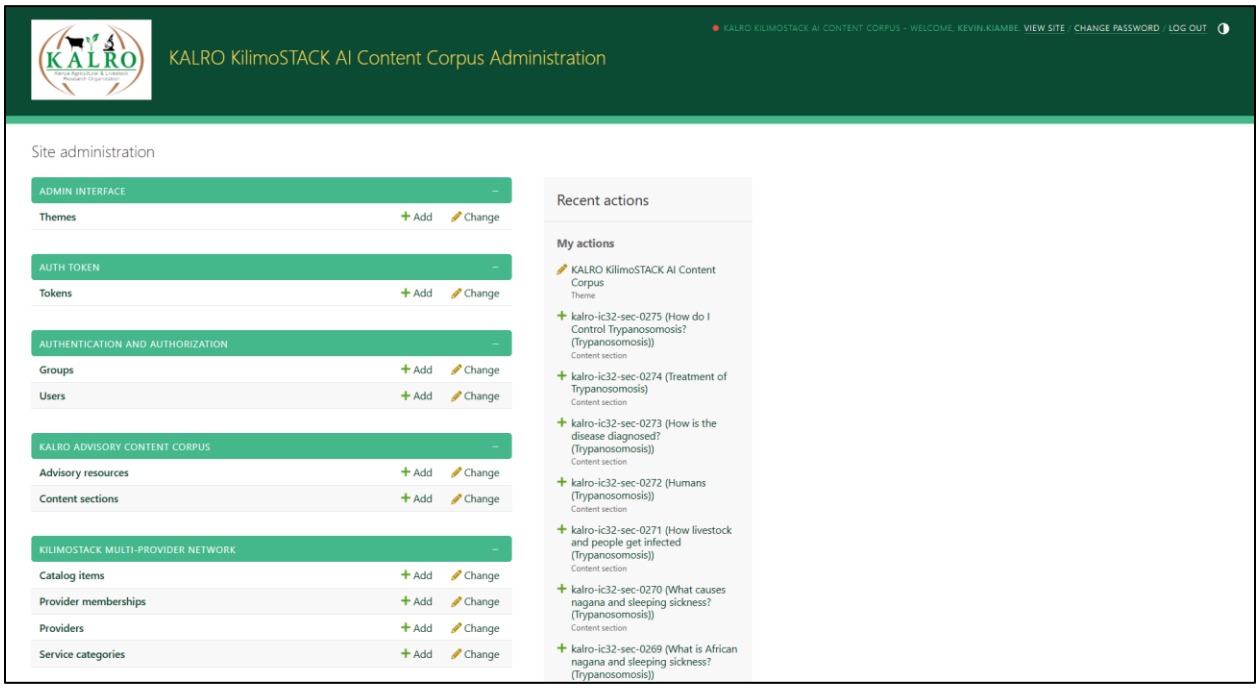
Expected Output	Description	Status	Evidence / Notes
Output 1	Harmonized national advisory content inventory across priority crop, soil, livestock, pest/disease and climate value chains	Achieved during the workshop	A 28-item value-chain inventory was compiled, screened and cross-checked against existing KALRO and partner repositories on Day 2, and formally signed off in plenary. Post-workshop, 5 of the 28 value chains (Indigenous Chicken, Beans, Green Grams, Natural Resource Management, and African Indigenous Vegetables) have been fully curated and uploaded to the platform.
Output 2	Agreed AI-ready metadata, AEZ-localization and content-classification standards	Drafted and pilot-tested	A metadata tagging and classification schema, AEZ-localization framework, seasonal/climate tagging conventions, and content certification criteria were drafted on Day 3 and piloted against sample inventory entries. Formal organization-wide sign-off should be confirmed as a follow-up action.

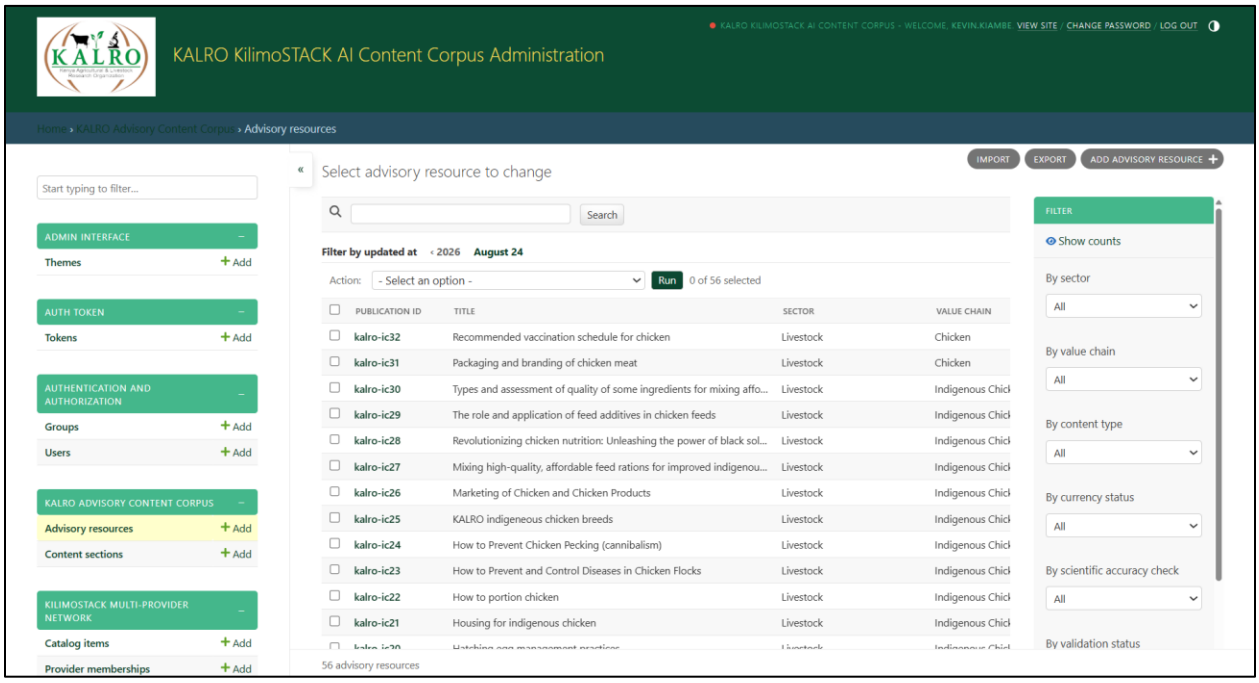
Output 3	Kenya AI Advisory Knowledge Hub design framework, including One Health content validation	Draft framework produced	Day 4 produced a draft Hub architecture, content taxonomy, federation model for external institutions, and governance/update-cycle model, alongside a completed review of the One Health advisory content set. The draft framework requires final validation and endorsement.
Output 4	Stakeholder-validated roadmap for content digitization and integration into KilimoSTACK and the national AI Advisory Platform	Presented for stakeholder validation	A digitization and integration roadmap, RACI matrix and indicative timeline were developed on Day 5 and presented in a stakeholder validation session, with feedback incorporated into a consolidated action plan closing the workshop.

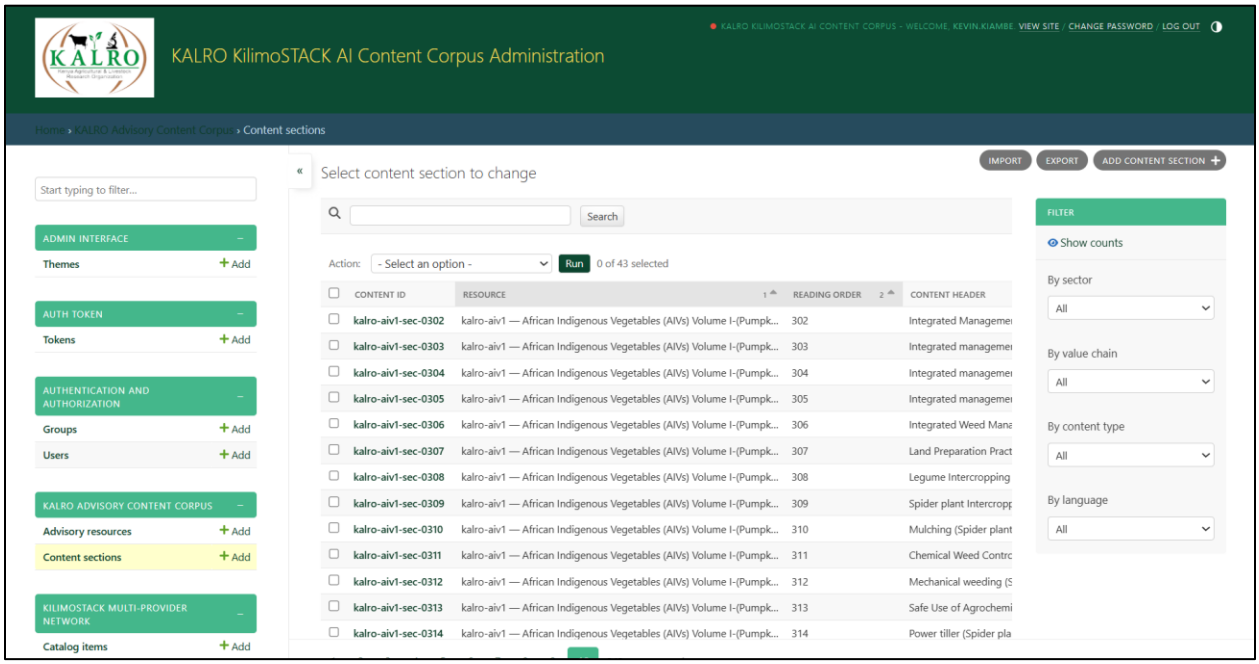
5. Recommendations and Way Forward

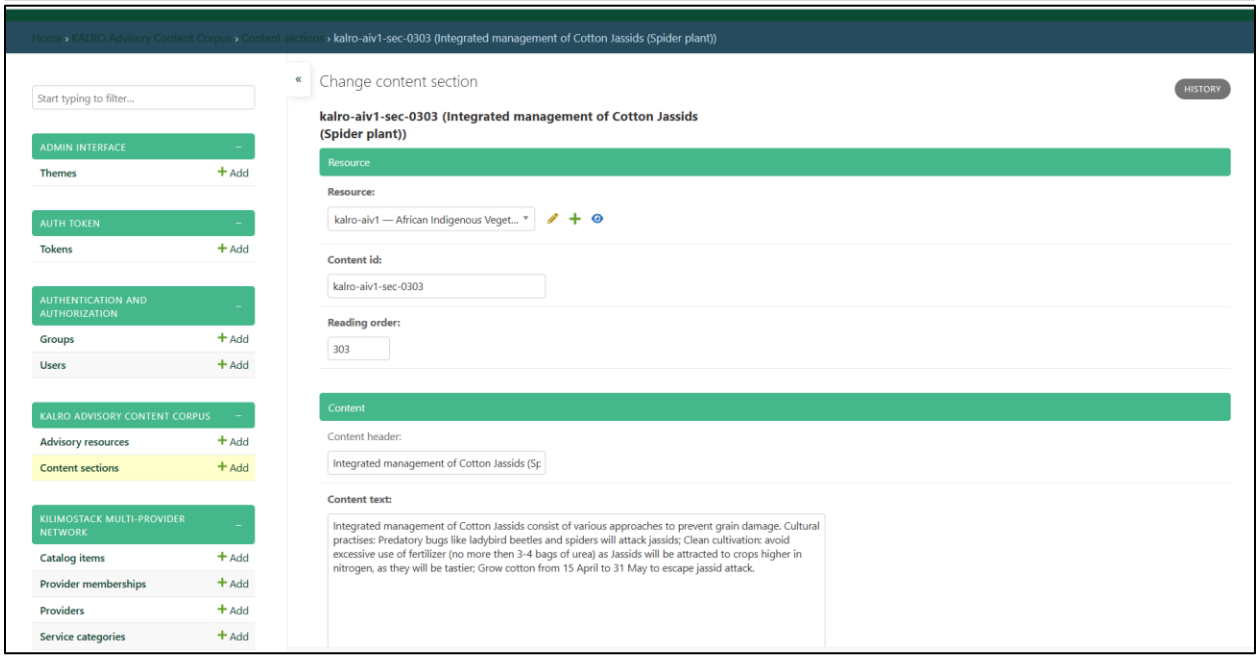
- Establish a KALRO Knowledge and Information Hub, as recommended by the Director, ICT & DA in plenary, so that curated advisory information is readily available organization-wide when required.
- Formally sign off the Day 3 metadata/AEZ/certification standards and the Day 4 Hub design framework, mirroring the plenary sign-off process already completed for Output 1.
- Schedule a follow-up review workshop or meeting (proposed within 4-6 weeks) to assess upload completion, confirm standards sign-off, and validate readiness for the next implementation phase.

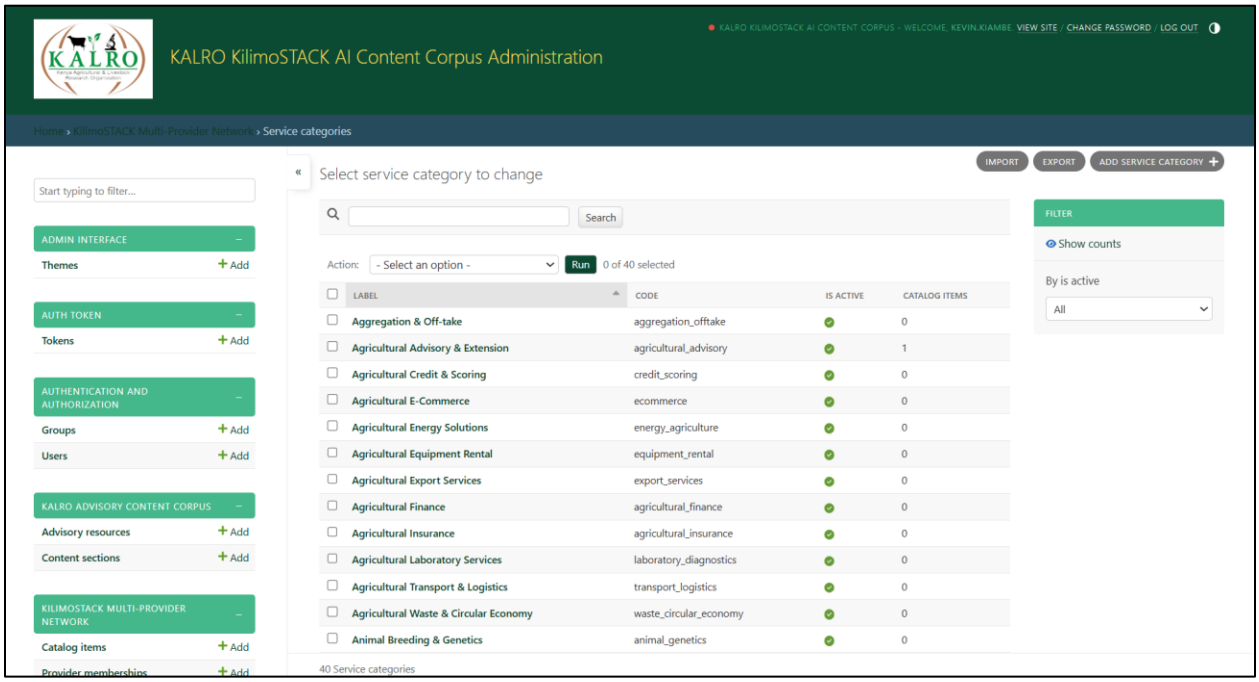
Appendix 1. Screenshots

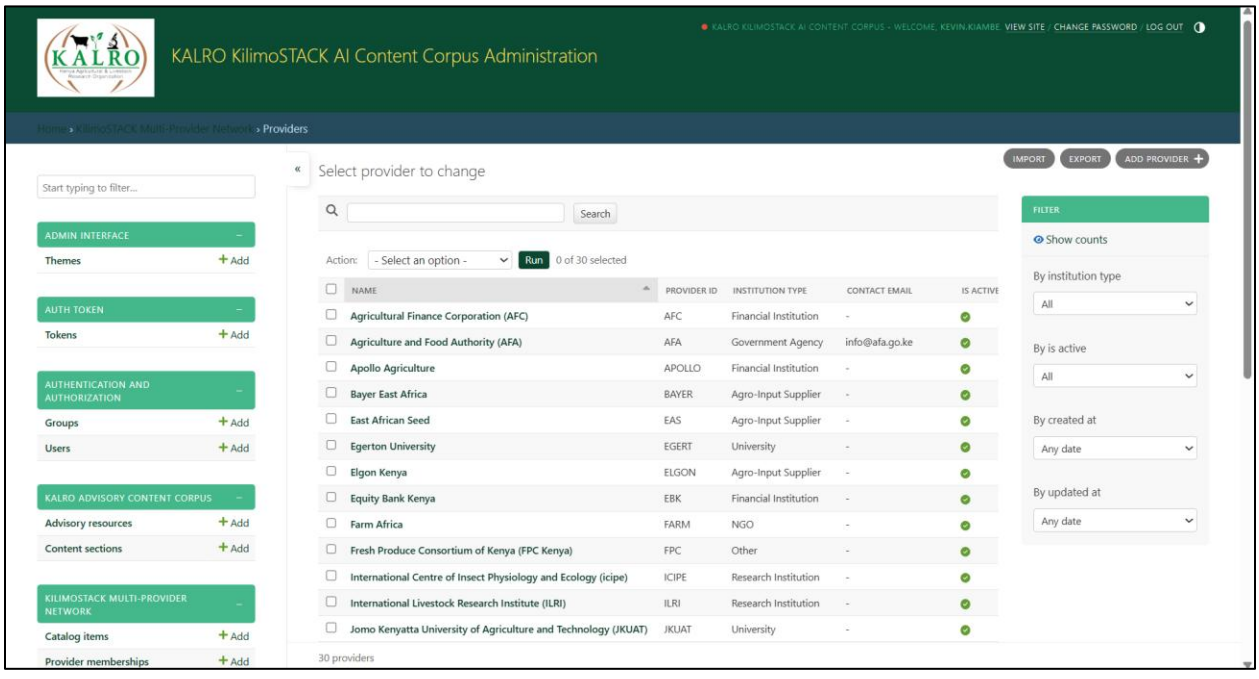


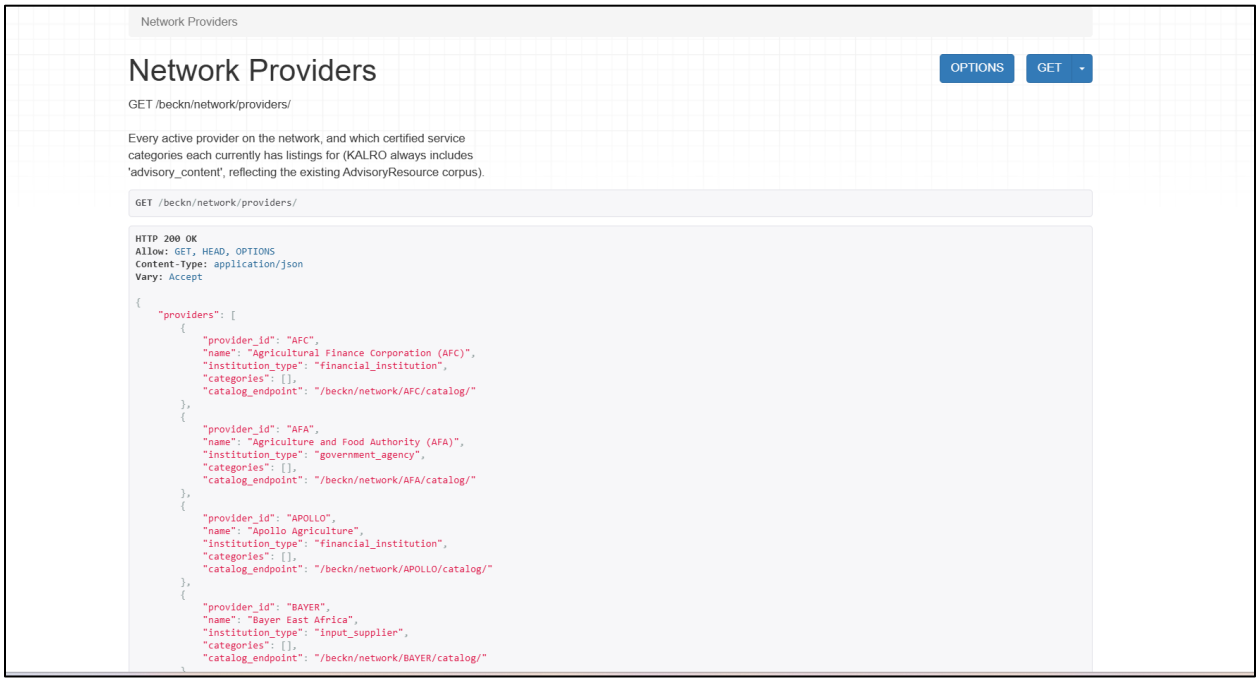



KALRO KilimoSTACK AI Content Corpus Administration
KALRO KILIMOSTACK AI CONTENT CORPUS - WELCOME, KEVIN.KIAMBE. VIEW SITE / CHANGE PASSWORD / LOG OUT
Home › KALRO Advisory Content Corpus › Content sections
Start typing to filter...
ADMIN INTERFACE
Themes + Add
AUTH TOKEN
Tokens + Add
AUTHENTICATION AND AUTHORIZATION
Groups + Add
Users + Add
KALRO ADVISORY CONTENT CORPUS
Advisory resources + Add
Content sections + Add
KILIMOSTACK MULTI-PROVIDER NETWORK
Catalog items + Add
Select content section to change
IMPORT
EXPORT
ADD CONTENT SECTION +
Search
Action: - Select an option - Run 0 of 43 selected
CONTENT ID
RESOURCE
READING ORDER
CONTENT HEADER
kalro-aiv1-sec-0302 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 302 Integrated Manageme
kalro-aiv1-sec-0303 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 303 Integrated manageme
kalro-aiv1-sec-0304 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 304 Integrated manageme
kalro-aiv1-sec-0305 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 305 Integrated manageme
kalro-aiv1-sec-0306 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 306 Integrated Weed Mana
kalro-aiv1-sec-0307 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 307 Land Preparation Pract
kalro-aiv1-sec-0308 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 308 Legume Intercropping
kalro-aiv1-sec-0309 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 309 Spider plant Intercropp
kalro-aiv1-sec-0310 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 310 Mulching (Spider plant
kalro-aiv1-sec-0311 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 311 Chemical Weed Contro
kalro-aiv1-sec-0312 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 312 Mechanical weeding (S
kalro-aiv1-sec-0313 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 313 Safe Use of Agrochemi
kalro-aiv1-sec-0314 kalro-aiv1 — African Indigenous Vegetables (AIVs) Volume I-(Pumpk... 314 Power tiller (Spider pla
FILTER
Show counts
By sector
All
By value chain
All
By content type
All
By language
All


Home › KALRO Advisory Content Corpus › Content sections › kalro-aiv1-sec-0303 (Integrated management of Cotton Jassids (Spider plant))
Start typing to filter...
ADMIN INTERFACE
Themes + Add
AUTH TOKEN
Tokens + Add
AUTHENTICATION AND AUTHORIZATION
Groups + Add
Users + Add
KALRO ADVISORY CONTENT CORPUS
Advisory resources + Add
Content sections + Add
KILIMOSTACK MULTI-PROVIDER NETWORK
Catalog items + Add
Provider memberships + Add
Providers + Add
Service categories + Add
Change content section
HISTORY
kalro-aiv1-sec-0303 (Integrated management of Cotton Jassids (Spider plant))
Resource
Resource:
kalro-aiv1 — African Indigenous Veget...
Content id:
kalro-aiv1-sec-0303
Reading order:
303
Content
Content header:
Integrated management of Cotton Jassids (Sp
Content text:
Integrated management of Cotton Jassids consist of various approaches to prevent grain damage. Cultural practises: Predatory bugs like ladybird beetles and spiders will attack jassids; Clean cultivation: avoid excessive use of fertilizer (no more then 3-4 bags of urea) as Jassids will be attracted to crops higher in nitrogen, as they will be tastier; Grow cotton from 15 April to 31 May to escape jassid attack.


KALRO KilimoSTACK AI Content Corpus Administration
KALRO KILIMOSTACK AI CONTENT CORPUS - WELCOME, KEVIN.KIAMBE. VIEW SITE / CHANGE PASSWORD / LOG OUT
Home › KilimoSTACK Multi-Provider Network › Service categories
IMPORT
EXPORT
ADD SERVICE CATEGORY
Start typing to filter...
ADMIN INTERFACE
Themes + Add
AUTH TOKEN
Tokens + Add
AUTHENTICATION AND AUTHORIZATION
Groups + Add
Users + Add
KALRO ADVISORY CONTENT CORPUS
Advisory resources + Add
Content sections + Add
KILIMOSTACK MULTI-PROVIDER NETWORK
Catalog items + Add
Provider memberships + Add
Select service category to change
Search
Action: - Select an option - Run 0 of 40 selected
LABEL
CODE
IS ACTIVE
CATALOG ITEMS
Aggregation & Off-take aggregation_offtake 0
Agricultural Advisory & Extension agricultural_advisory 1
Agricultural Credit & Scoring credit_scoring 0
Agricultural E-Commerce ecommerce 0
Agricultural Energy Solutions energy_agriculture 0
Agricultural Equipment Rental equipment_rental 0
Agricultural Export Services export_services 0
Agricultural Finance agricultural_finance 0
Agricultural Insurance agricultural_insurance 0
Agricultural Laboratory Services laboratory_diagnostics 0
Agricultural Transport & Logistics transport_logistics 0
Agricultural Waste & Circular Economy waste_circular_economy 0
Animal Breeding & Genetics animal_genetics 0
40 Service categories
FILTER
Show counts
By is active
All


KALRO KilimoSTACK AI Content Corpus Administration
KALRO KILIMOSTACK AI CONTENT CORPUS - WELCOME, KEVIN.KIAMBE. VIEW SITE / CHANGE PASSWORD / LOG OUT
Home › KilimoSTACK Multi-Provider Network › Providers
IMPORT
EXPORT
ADD PROVIDER
Start typing to filter...
ADMIN INTERFACE
Themes + Add
AUTH TOKEN
Tokens + Add
AUTHENTICATION AND AUTHORIZATION
Groups + Add
Users + Add
KALRO ADVISORY CONTENT CORPUS
Advisory resources + Add
Content sections + Add
KILIMOSTACK MULTI-PROVIDER NETWORK
Catalog items + Add
Provider memberships + Add
Select provider to change
Search
Action: - Select an option - Run 0 of 30 selected
NAME
PROVIDER ID
INSTITUTION TYPE
CONTACT EMAIL
IS ACTIVE
Agricultural Finance Corporation (AFC) AFC Financial Institution -
Agriculture and Food Authority (AFA) AFA Government Agency info@afa.go.ke
Apollo Agriculture APOLLO Financial Institution -
Bayer East Africa BAYER Agro-Input Supplier -
East African Seed EAS Agro-Input Supplier -
Egerton University EGERT University -
Elgon Kenya ELGON Agro-Input Supplier -
Equity Bank Kenya EBK Financial Institution -
Farm Africa FARM NGO -
Fresh Produce Consortium of Kenya (FPC Kenya) FPC Other -
International Centre of Insect Physiology and Ecology (icipe) ICIPE Research Institution -
International Livestock Research Institute (ILRI) ILRI Research Institution -
Jomo Kenyatta University of Agriculture and Technology (JKUAT) JKUAT University -
30 providers
FILTER
Show counts
By institution type
All
By is active
All
By created at
Any date
By updated at
Any date



Network Providers

Network Providers

OPTIONS GET

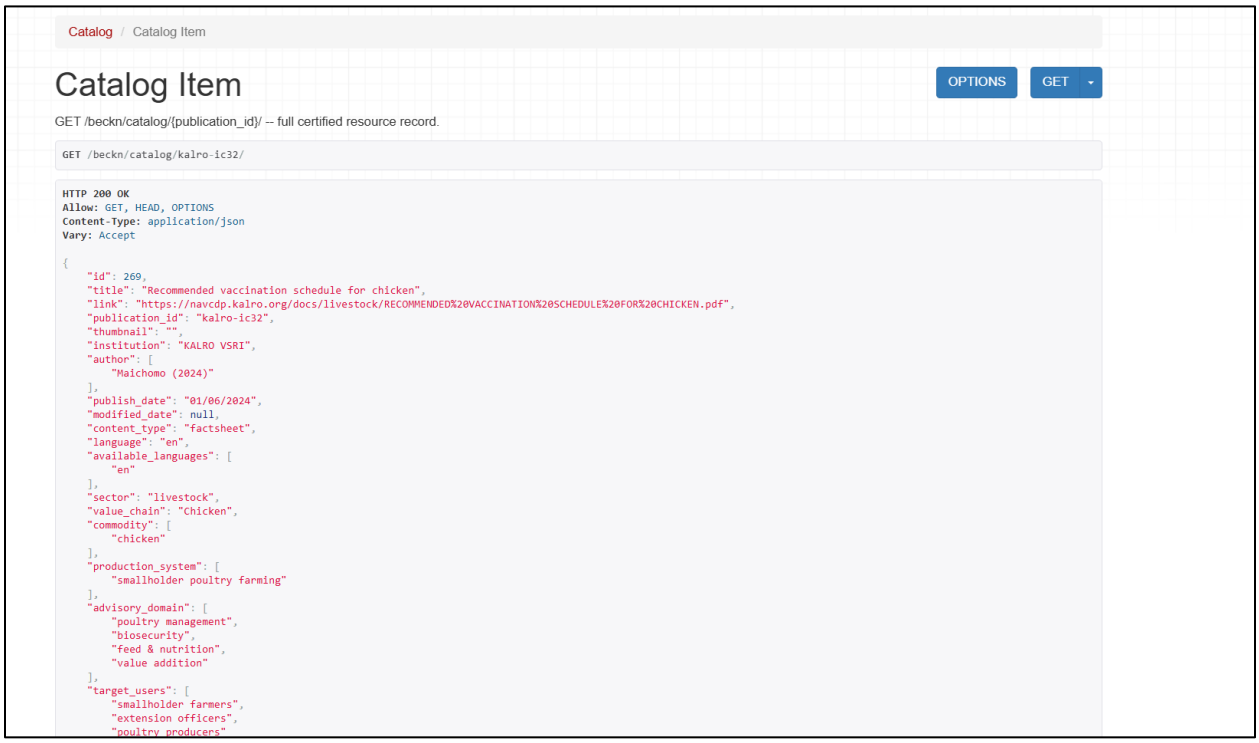
GET /beckn/network/providers/

Every active provider on the network, and which certified service categories each currently has listings for (KALRO always includes 'advisory_content', reflecting the existing AdvisoryResource corpus).

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GET /beckn/network/providers/
```

```
HTTP 200 OK
Allow: GET, HEAD, OPTIONS
Content-Type: application/json
Vary: Accept

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        {
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        },
        {
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        },
        {
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```



Catalog / Catalog Item

Catalog Item

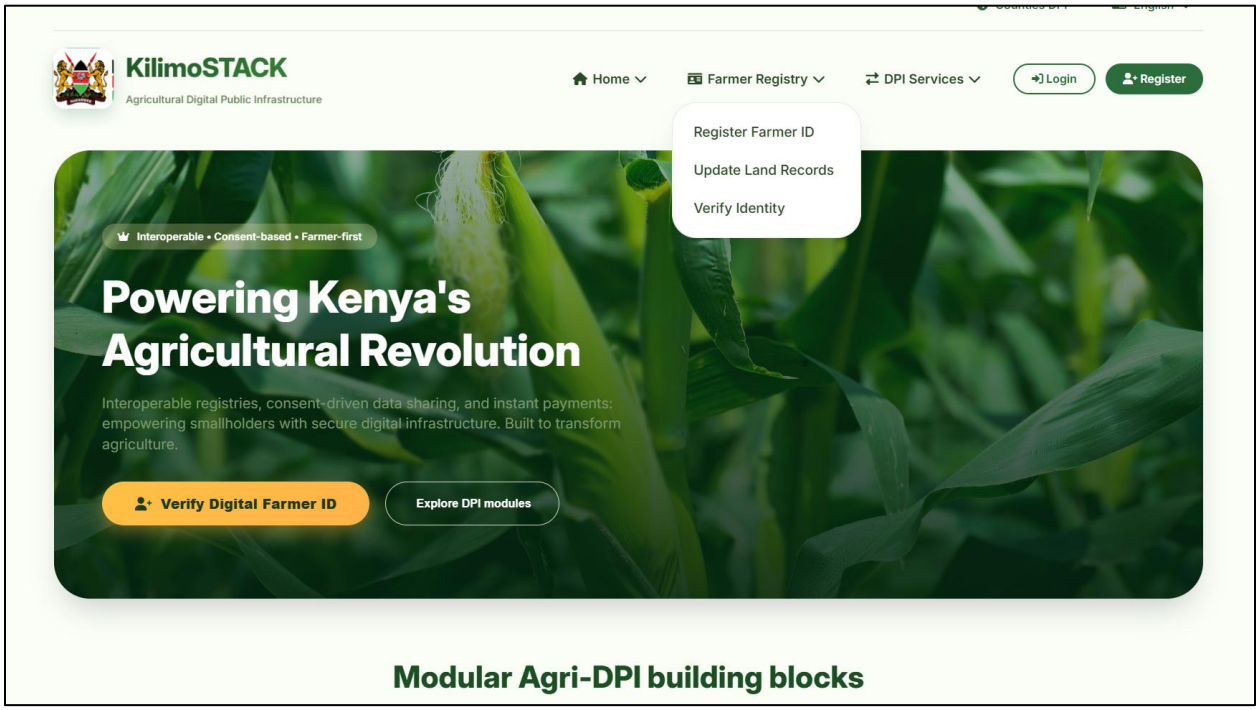
OPTIONS GET

GET /beckn/catalog/{publication_id}/ -- full certified resource record.

```
GET /beckn/catalog/kalro-ic32/
```

```
HTTP 200 OK
Allow: GET, HEAD, OPTIONS
Content-Type: application/json
Vary: Accept

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        "biosecurity",
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        "extension officers",
        "poultry producers"
```



Modular Agri-DPI building blocks